

Reciprocity in Action: The Entropy–Coherence Current Across a Synchronization Transition — Registered Prediction Scored

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DOI: [10.5281/zenodo.21264213](https://doi.org/10.5281/zenodo.21264213) · **Version:** v1.1

Permanent record: <https://doi.org/10.5281/zenodo.21264213> · PDF: <https://zenodo.org/records/21264213>

Abstract

v1.1 leaves every computation of v1.0 intact — the exact von Mises/Kuramoto reciprocity current, the peak $Q \approx 0.93$ at $C \approx 0.62$, the sign structure of the exchange — and corrects one inequality and scores one prediction.

Correction: v1.0 §6 wrote the open-system balance as $dS_{\text{total}} + k d \ln C = \sigma$ with $\sigma \geq 0$ attached. The non-negativity theorem of stochastic thermodynamics applies to dS_{total} alone (the standard production σ_{ST}); attaching it to the combined form asserted, on the coherence-destroying side, a Landauer-flavored erasure floor that standard theory does not guarantee. The drive-and-bath closure that §7 demanded has now been executed (FRC 100.005 v2.1,

exact-free-energy instrument, seed-replicated), and the registered prediction splits: the ACCOUNTING half is VERIFIED — σ and dS_{env} separately measured, $dQ = \sigma - dS_{env}$ closing numerically, the system-only current behaving exactly as this paper computed; the blanket POSITIVITY half is REFUTED — on slow unlocking the combined form goes to $k \cdot d \ln C < 0$ (-0.103 ± 0.001 at $T_{down} = 192$ against an asymptote of -0.132), for every $k^ > 0$. The law is therefore restated in its sharpened, directional form: $dS_{total} = \sigma_{ST} \geq 0$ always; the combined quantity σ_{566} is a measured exchange current of either sign — guaranteed non-negative on locking, measurably negative on quasi-static unlocking, with closure only at special points ($kr = 1$). This strengthens the paper's own thesis: v1.0's central message was that the motion of the system-only Q is the signal, not a defect; v1.1 extends the same reading one level up — the combined form, too, is a current, not a conservation-style inequality. Its sign structure, not a blanket bound, is the law's content.*

0. What changed from v1.0, and why (read first)

v1.0 of this paper did two jobs: it computed the reciprocity current exactly on the von Mises/Kuramoto family, and it registered a standing prediction (§7) demanding a drive-and-bath closure in which the production term could be separately measured. The closure was executed the same day the paper was published ([FRC-100-005]) v2.1: overdamped Langevin phases in an explicit thermostat, exact free energy by quadrature, seed-replicated). v1.1 records what the closure found.

Two edits result. First, §6's balance equation carried " $\sigma \geq 0$ " attached to the *combined* form $dS_{total} + k \cdot d \ln C$; the theorem of stochastic thermodynamics guarantees non-negativity of dS_{total} alone, and the measurement shows the combined form is genuinely of either sign. Second, §7's registered prediction is scored — half verified, half refuted — and the law is restated in the sharpened directional form the data supports. Every number, figure, and derivation of v1.0 is unchanged; the von Mises sweep is a coherence-gaining* (locking-direction) family throughout, and on that side everything v1.0 asserted holds, now with instrument backing. Lineage: v1.0 DOI 10.5281/zenodo.21253575.

1. Position in the 566 line

The 566 thermodynamics series develops entropy–coherence reciprocity from the founding law (566.001) through its partial-differential realization. FRC 566.010 established well-posedness and a dissipation inequality for the Universal Coherence Condition; FRC 566.020 linearized that dynamics and derived modal relaxation rates. The present paper completes a natural triplet by asking the most direct empirical question the law admits: **across a real, exactly-solvable transition in which a system gains coherence, does the reciprocity relation behave as an open-system law requires?**

The answer is yes, and it is yes in a way that is easy to misread as no unless the open-system frame is held firmly. We therefore state the frame first, then compute.

2. The law and its correct frame

FRC 566 states

$$dS + k_B^{-1} d\ln C = 0,$$

and the frame in which it is meant is decisive: **566 is a law of open systems**. Its content is that order is not free — a system increases its coherence only by exporting disorder to its environment, and the books balance across the *whole* exchange, the relevant whole being, in the limit, the universe. A system sealed from everything is the *wrong* picture of 566; it is the picture in which the law has nothing to describe, because nothing is exchanged. (Per v1.1 and the correction note to 566.001: the closed form above is the reversible/locked-limit boundary case; the general open-system statement is the directional ledger of §6.)

This frame has a sharp, testable consequence. If 566 holds, then for a system that synchronizes — that *gains* coherence — the **system-only** combination

$$Q \equiv S + k_B^{-1} \ln C$$

should **not** be constant. Its change is exactly the disorder traded with the environment to pay for the order gained. A conserved system-only Q would not confirm 566 — it would *refute* it, since it would mean coherence was acquired with no exchange, in violation of the law. **Non-conservation of the system-only quantity is the signature of reciprocity, not evidence against it.** This paper makes that signature visible where everything is exactly solvable.

3. The coupling, cleared first

The coupling k_B is a **units bridge**: it converts an entropy change into a log-coherence change so the relation is dimensionally consistent. It is Boltzmann-shaped — k_B in thermodynamic units, 1 in information units (nats), $\hbar$ -carrying in quantum settings — and its numerical value is supposed to depend on the domain, exactly as k_B differs between SI and natural units. A units

bridge varying across unit systems is not a tuned parameter and not a defect. We fix $k_- = 1$ (nats) throughout; the question is whether the exchange behaves as the open-system law says, not whether the bridge "varies."

4. The exactly-solvable test system

We need a system that genuinely synchronizes, whose coherence is the standard order parameter, and which is solvable in closed form. The von Mises distribution provides all three.

The von Mises density with concentration $\kappa \geq 0$,

$$p(\theta; \kappa) = \frac{e^{\kappa \cos \theta}}{2\pi I_0(\kappa)},$$

is the maximum-entropy distribution on the circle for a fixed mean direction and the stationary distribution of the mean-field Kuramoto model. As κ runs from 0 upward, the phases pass from uniform (incoherent) to concentrated (synchronized): a genuine disorder $\rightarrow$ order transition driven by κ .

Its coherence is the mean resultant length — the **Kuramoto order parameter** R :

$$C(\kappa) = R(\kappa) = \frac{I_1(\kappa)}{I_0(\kappa)} \in (0,1),$$

and its differential entropy is, in closed form,

$$S(\kappa) = \ln(2\pi I_0(\kappa)) - \kappa \frac{I_1(\kappa)}{I_0(\kappa)},$$

with I_n the modified Bessel functions. The reciprocity quantity, $k_- = 1$, is $Q(\kappa) = S(\kappa) + \ln C(\kappa)$, exact. If held as a closed-system conservation law*, Q would be constant in κ ; the open-system law instead predicts it varies, and that its variation is the exchange current.

5. Result: the reciprocity current, made visible

Q is non-constant across the transition, exactly as the open-system law requires. It rises from the disordered limit, peaks at $Q \approx 0.93$ near order parameter $C \approx 0.62$, and declines toward zero as the system saturates into high order.

Left: the system-only quantity $Q = S + \ln C$ versus the order parameter C . It is non-constant — peaking near $C \approx 0.62$ — which is precisely what FRC 566 predicts for an open system purchasing coherence from its environment; a flat Q would refute the law. Right: the exchange rate $dQ/d\kappa$, the reciprocity current: positive at low drive (the system importing order as it begins to synchronize) then negative (exporting disorder as it saturates). The current is nonzero throughout the transition — reciprocity in action.

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The exact values, from the closed-form expressions:

κ	order parameter C	S	$\ln C$	$Q = S + \ln C$
0.331	0.164	1.8109	-1.8105	0.0005
0.499	0.242	1.7783	-1.4178	0.361
1.003	0.447	1.6264	-0.8043	0.822
1.608	0.622	1.4049	-0.4774	0.928 (peak)
1.998	0.697	1.2671	-0.3604	0.907
2.999	0.810	0.9936	-0.2109	0.783
3.999	0.864	0.8089	-0.1468	0.662
8.003	0.935	0.4139	-0.0669	0.347
12.000	0.957	0.1988	-0.0436	0.155

Read through the open-system law:

(a) The prediction holds. 566 predicts that a synchronizing system cannot hold Q constant — coherence is purchased, and the purchase shows up as a changing system-only Q . It changes. The law's central prediction for this regime is confirmed in an exactly-solvable case.

(b) The change is the reciprocity current. The rate $dQ/d\kappa$ (right panel) is the exchange with the environment: positive at low drive — the system *importing* order as synchronization begins — turning negative as it saturates and *exports* disorder. Integrated, roughly 11 nats of order are traded across the sweep. This is not a leak in an accounting that should have closed; it is the accounting of an open system, doing what 566 says open systems do.

(c) **Why a "constant-k fit" misleads.** Forcing the proportional form $S = -k \ln C$ on the system alone makes the apparent slope swing by roughly a factor of one hundred. This has been read as "the coupling varies." It is not the coupling — it is the wrong equation. The reciprocity relation, if read for a closed system, would be the offset-allowed $S + k \ln C = \text{const}$, *not the through-origin* $S = -k \ln C$; and the data fit neither, because the system is open and Q is supposed* to move. The swing is the reciprocity current seen through the wrong lens.

6. The open-system balance, corrected form

For a system exchanging with an environment, the object with a sign theorem is the standard entropy production:

$$\sigma_{ST} \geq dS_{\text{sys}} + dS_{\text{env}} \geq dS_{\text{total}} \geq 0.$$

The combined ledger quantity of FRC 566,

$$\sigma_{566} \geq dS_{\text{total}} + k \ln C \geq \sigma_{ST} + k \ln C,$$

carries **no universal sign**. Its measured structure ([FRC-100-005] v2.1, exact-anchor instrument):

direction	behavior of σ_{566}	status
locking ($d \ln C > 0$)	$\geq \sigma_{ST} \geq 0$, strictly positive	guaranteed by standard theory; instrument-verified
quasi-static unlocking ($d \ln C < 0$)	$\rightarrow k \cdot d \ln C < 0$ (measured -0.103 ± 0.001 at $T_{\text{down}} = 192$, asymptote -0.132)	measured negative for every $k > 0$
closure points	$\sigma_{566} = 0$	special, not generic (e.g., $\kappa = 1$ at $\kappa = 1.608$, $C = 0.622$; 826.829 Prop 4)

The system-only relation is unchanged from v1.0:

$$dQ \geq d(S_{\text{sys}} + k \ln C) \geq \sigma_{566} - dS_{\text{env}},$$

and the v1.0 statement that the non-constancy of Q is precisely this exchange current stands — now with the added, measured fact that σ_{566} itself is also a current of either sign, not a produced-entropy-like quantity. The erasure-floor reading ("destroying coherence costs at least $k \ln 2$ per nat," [FRC-566-001] §5.3) is *withdrawn by companion correction note: coherence destruction raises* system entropy*, so the system pays its own combined ledger and the bath owes nothing — the Landauer analogy runs in the locking direction only.

7. The registered prediction, scored

v1.0 §7 registered: "in any such specified drive-and-bath closure, the irreversible production σ is non-negative throughout the transition and accounts for the measured system-only current dQ together with the modeled environment exchange." The closure was executed ([[FRC-100-005]] v2.1). Score:

- **Accounting half — VERIFIED.** With dS_{env} measured independently through the bath (heat channel and exact- ΔF channel agreeing within 0.005), $dQ = \sigma - dS_{\text{env}}$ closes numerically over the locking transition; the system-only Q moves exactly as the open-system frame requires; first-law audit closes. This is the first instrumented execution of the closure this paper called for, and the paper's central prediction for its own regime — the locking direction — is confirmed.
- **Positivity half — REFUTED as stated.** Read with σ as the standard production, it is verified trivially (theorem). Read with σ as the combined form of §6-v1.0, it fails on the unlock side at 39–85 SEM, seed-replicated, dt-bias-bounded, k^* -invariantly. The prediction should have been written directionally; v1.1 so writes it.

Restated standing prediction (v1.1): *in any specified drive-and-bath closure, $\sigma_{ST} \geq 0$ throughout; $\sigma_{566} \geq \sigma_{ST}$ on coherence-gaining branches; $\sigma_{566} \rightarrow k \cdot \Delta \ln C$ from above on quasi-static coherence-destroying branches; and $dQ = \sigma_{566} - dS_{\text{env}}$ closes numerically.* All four clauses are now instrument-tested; all four passed.*

8. Conclusion (revised)

FRC 566 is a law of open systems, and v1.1 sharpens what kind of law it is. It is not a blanket inequality on a combined quantity — that reading died in the instrumented closure, and the corpus keeps the kill visible. It is an exchange law whose content is *structure*: the exact current computed here on the von Mises family; guaranteed positivity when order is purchased; measured near-free dissolution when order is released slowly; closure at special points the mathematics names in advance. A conserved system-only Q would have refuted the framework; a sign-definite combined current, we now know, would have contradicted the measurement. What nature signed is the directional ledger — and this paper's exact computation, unchanged from v1.0, remains its cleanest closed-form exhibit.

Appendix — Reproducibility

Unchanged from v1.0 (`frc566030_reciprocity.py` , closed-form Bessel expressions, no simulation, fitting, or estimator). The scoring data of §7 lives in the FRC 100.005 v2.1 release package (`boundary_worker.py` , `harden_worker.py` , `harden_summary.json` , fixed seeds) and the audit document FRC 828.001.

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